

SAI VAMSI SAHUKARI

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EXPERIENCE

Arista Networks

Software Engineer (July 2023 – Present)

Unified Campus Fabric Operations

- Architected SWAG (Switch Aggregation Group), a campus fabric architecture that unifies up to 16 switches into a single logical system on hardware supporting 8 GB supervisor memory.
- Designed and automated end-to-end provisioning workflows for campus network deployments, scaling to 2,000+ devices and 250,000+ interfaces across distributed switch topologies, with 4 enterprise clients actively onboarded on telemetry.
- Provided technical mentorship to 2+ engineers on system architecture, testing practices, and codebase navigation, helping new hires become productive quickly.
- Developed a CloudVision Portal app to provision SWAG devices, leveraging Configuration Studio to automate end-to-end config generation and deployment across 24, 48, and 96-interface SKUs for flexible campus network topologies.

CloudVision Pathfinder — SD-WAN Development

- Spearheaded CloudVision Pathfinder (Arista's SD-WAN solution) development from design through CVaaS GA, delivering provisioning, telemetry, and visibility dashboards that enabled adoption by 3 enterprise clients.
- Built and maintained 50+ Go-based microservices processing telemetry from 1,000+ routers, storing versioned data in HBase with sub-second latency for real-time visibility.
- Monitored production system health via Grafana dashboards tracking 25+ alert categories and 100+ metrics across turbine lifecycle, memory, CPU, and gRPC layers, enabling rapid detection and resolution of critical issues such as OOM kills, panics, and data loss events.
- Debugged and resolved 30+ complex issues across features, ensuring reliability and performance in production environments.
- Optimized turbine pod restart time from 50 minutes to 2 minutes (96% improvement) by refactoring the subscription pipeline for high-notification input paths.

EDUCATION

National Institute of Technology, Warangal

GPA: 7.95/10

B.Tech in Computer Science and Engineering (2019–2023)

PROJECTS

Food Delivery Application [GitHub]

- Developed **15+ secure RESTful APIs** using **Java Spring Boot** with **JWT-based stateless authentication**, implementing selective endpoint authorization — segregating public and protected routes for admin and user operations using **Spring Security**.
- Integrated **AWS S3** for cloud-based food image storage and configured **MongoDB** to persist image URLs, alongside **BCrypt password encryption** and a custom **JWT filter chain** for token validation on every request.

Software Effort Estimation using Multi-objective Binary Rao Algorithm 📄

- Published IEEE-indexed research proposing a Multi-Objective Binary Rao Algorithm for software effort estimation, outperforming NSGA-II and Single Objective Binary Rao across **4** benchmark PROMISE datasets (COCOMO, China, Albrecht, Kemerer) totaling 591 project instances.
- Engineered a feature selection pipeline reducing optimal features from up to 19 down to 1–7 per dataset, evaluated across 3 ML models (Linear Regression, SVR, Random Forest) achieving improved MAE and RMSE scores over baseline algorithms.

TECHNICAL SKILLS

- **Languages:** Go, Java, C++, C, Python
- **Frameworks & API Design:** Spring Boot, gRPC, REST APIs
- **Databases:** MySQL, Apache HBase, MongoDB
- **Messaging & Event Streaming:** Apache Kafka
- **Cloud & DevOps:** Spinnaker, Docker, CI/CD Pipelines, Kubernetes, AWS S3
- **Observability & Monitoring:** Grafana, Prometheus
- **Core Competencies:** System Design, Microservices Architecture, Data Structures and Algorithms, Computer Networks, Production Debugging, Technical Leadership